Machine Learning Basics

Machine learning is a field of artificial intelligence that enables computers to learn patterns from data without being explicitly programmed.

Types of Machine Learning

Supervised learning uses labeled data to train predictive models.

Unsupervised learning discovers hidden structures and patterns in unlabeled datasets.

Reinforcement learning allows an agent to learn through interaction with an environment.

Model Training

Training involves minimizing a loss function using optimization algorithms such as gradient descent.

Model quality depends heavily on the quality of training data and feature engineering.

Evaluation

Common evaluation metrics include:

Accuracy

Precision

Recall

F1 Score

ROC-AUC

Cross-validation helps estimate generalization performance.